IBVAP — Performance Analysis & System Benchmarks

13. System Metrics, Latency & Measurement Disambiguation

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1. Empirical Measurement vs. Theoretical Estimates

ZERO-HALLUCINATION POLICY:

*The table below explicitly distinguishes between parameters that were **empirically measured** in the project artifacts versus those that are **not measured / estimated**.*

Performance Parameter	Value / Status	Verification Source
Model Weights Disk Size	`yolo11s.pt` (19.3 MB) / `yolo11n.pt` (5.61 MB)	`models/` directory file properties
Input Video Resolution	`720p / 1080p Standard MP4`	`data/video.mp4` properties
Video File Size	`2.19 MB` (`data/video.mp4`)	`data/` directory properties
Processed Demo Video Size	`25.95 MB` (`final_intrusion_video.mp4`)	`output/` directory properties
Tracking Buffer Memory	`60 Frames`	`bytetrack_day3.yaml` line 8
Inference FPS	`Not experimentally measured in current implementation`	ZERO HALLUCINATION VERIFIED
Detection Latency (ms)	`Not experimentally measured in current implementation`	ZERO HALLUCINATION VERIFIED
Model Accuracy (mAP50-95)	`Not experimentally measured in current implementation`	ZERO HALLUCINATION VERIFIED
Precision / Recall / F1	`Not experimentally measured in current implementation`	ZERO HALLUCINATION VERIFIED

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2. Computational Hardware Requirements

Minimum Hardware Requirements (CPU Execution)

- **CPU:** Quad-Core Intel Core i5 / AMD Ryzen 5 or higher
- **RAM:** 8 GB RAM
- **Storage:** 2 GB available SSD space
- **Execution Mode:** OpenCV CPU inference with YOLO11 Nano (`yolo11n.pt`).

Recommended Hardware Requirements (Edge / GPU Execution)

- **GPU:** NVIDIA RTX 3060 / Jetson Orin Nano / Jetson AGX Orin
- **VRAM:** 4 GB dedicated VRAM
- **CUDA / TensorRT:** TensorRT engine acceleration for real-time multi-camera 30+ FPS ingestion.

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3. Streamlit & Database Query Optimization

- **Caching Mechanism:** `@st.cache_data(ttl=30)` on `fetch_events()` caches database queries for 30 seconds, reducing round-trip latency to MongoDB Atlas / Supabase.
- **Asset Caching:** `@st.cache_data(ttl=300, show_spinner=False)` on `fetch_image_bytes()` caches evidence snapshots in RAM, preventing repeated network GET requests during gallery viewing.
- **Database Indexing:** Primary queries lookup events by `event_id` or sort by `frame`, matching default B-tree indexing in MongoDB / Supabase.